Health-a-thon 2026

FINAL SUBMISSION

Sahara

A phone app for people with a serious illness, like advanced cancer, and their families. Their doctor discusses their end of life wishes; Sahara creates a clinical document, and an emergency doctor can read it in one scan.

TEAM NAME Zeros and Ones **TEAM CODE** 37MYUP

Shaik Abdul Basith · Shaik Muhammad Awaiz · Dr Sharada Vinod Kutty · Dr Sujay Halkur Shankar · Shirin Ayub

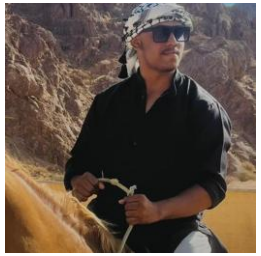
Try it: sahara.wedevit.in · Android app and website · English, Hindi, Telugu, Urdu, Kannada

Team Overview & Background



Shaik Abdul Basith
Technical lead
WeDevIT

Led the build of the website and Android app; the AI that transcribes the conversation and fills the goals-of-care document; answers from reports; server, security and releases



Shaik Muhammad Awaiz
Android engineer
WeDevIT

Built Android screens: patient summary, care timeline, doctor schedule, family invites; AI retry and backup models; tested every version



Dr Sharada Vinod Kutty
Pulmonologist
AIIMS Mangalagiri

Wrote the clinical specifications and designed the goals of care document. Tested mock clinical conversations



Dr Sujay Halkur Shankar
Pulmonologist
Tata IISc Medical School

Defined the problem statement and the proposed solution. Provided relevant clinical documents and scores for design of tools



Shirin Ayub
Medical student

Designed the symptom log and defined caregiver portal needs. Instrumental in UI clean up and usability inputs

Our doctors chose this problem from their own clinics and checked every version.

Legal advice was sought: inputs and refining the goals of care wording and format.

Built by WeDevIT, the team's company. The app is on Google Play under WeDevIT.

Problem Statement and Supporting Evidence



Example story (illustration). A 62-year-old woman with advanced breast cancer.

At the clinic, her doctor asks what treatment she would want if she got very sick. Only her husband is there.

Two months later she is unconscious in the emergency room. Should doctors try to restart her heart (CPR)?

Her children disagree, and nobody can find what she said.

98%

of India's metastatic (advanced) cancer patients had unmet palliative-care needs.

Patil, BMJ Support Palliat Care 2024

30–37%

of patients get timely palliative care referrals (two centres).

Venketeswaran, BMJ Support Palliat Care 2026

41%

of cancer patients dying on active treatment had end-of-life care documented (one hospital).

Das, Cancer Med 2026

88%

of urban adults want to choose their own end-of-life care.

Living Will Survey, IndiaSpend 2019

What has to change

Her goals of care need to be documented, signed, and found in seconds in an emergency.

Target Users and Use Case



Patient · at home
Logs symptoms, medications, and helps find care; access to clinical documents

Family · at home
Access to their patient's portal; logs specific caregiver issues

Their doctor · clinic
Guides goals of care discussions; access to patient information: symptom trends, reports, medications

Emergency doctor · emergency room
Quick retrieval of the latest goals of care discussion to guide emergency medical decisions

Words used in this deck

Palliative care: care that eases pain and distress in a serious illness.

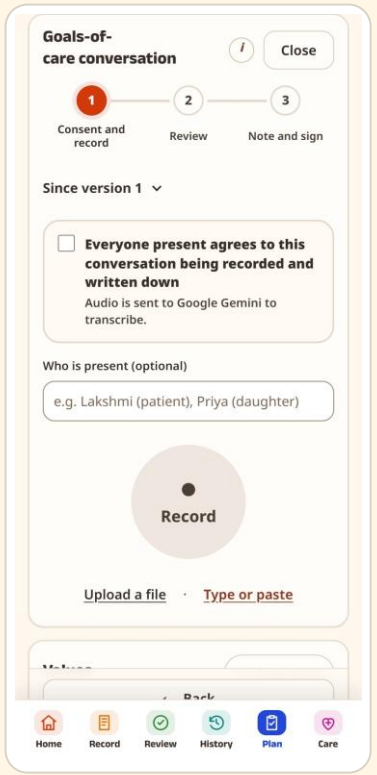
Goals of care: what treatment a patient wants, or refuses, near the end of life (CPR, ventilator, ICU).

QR code: a square barcode on the patient's phone or card; a phone camera scan opens their wishes.

Solution Overview, Key Features & Impact (1/2)

In the clinic: the doctor talks, Sahara listens and writes.

After consent, the doctor leads the goals-of-care conversation and records it on the doctor portal. Today this is rarely documented.



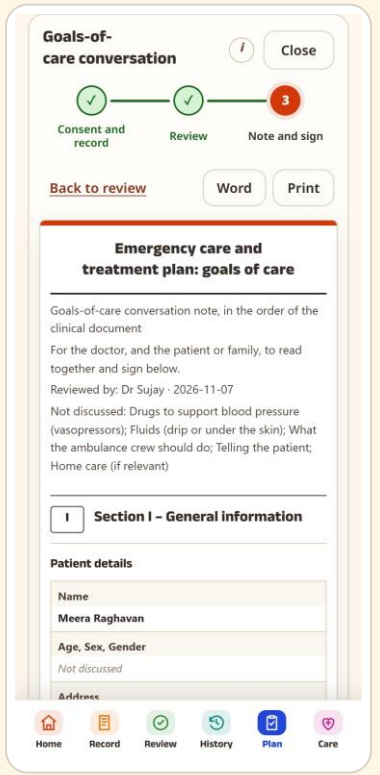
1. Patient agrees, tap Record

Google's Gemini AI types out who said what



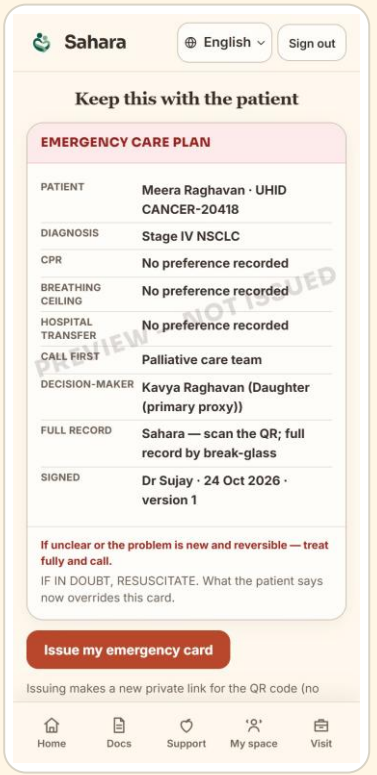
2. A checklist ticks itself

Missed topics are flagged. Android: Dr Sujay's 13 topics, with "Guide me" or "Free flow" taps



3. Answers fill the structured clinical document

Doctor checks the transcription and summary



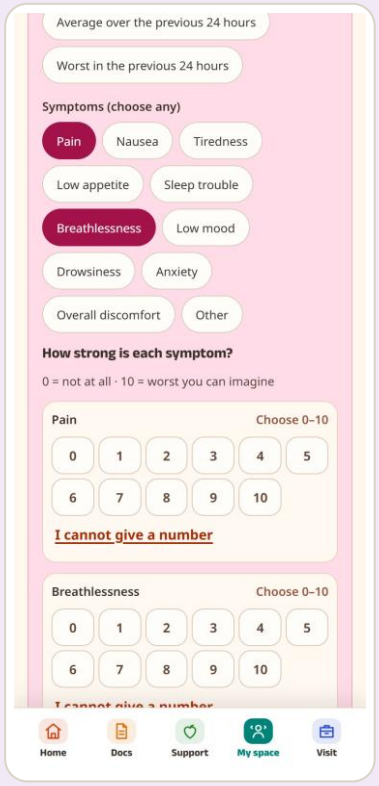
4. Everyone signs on the phone

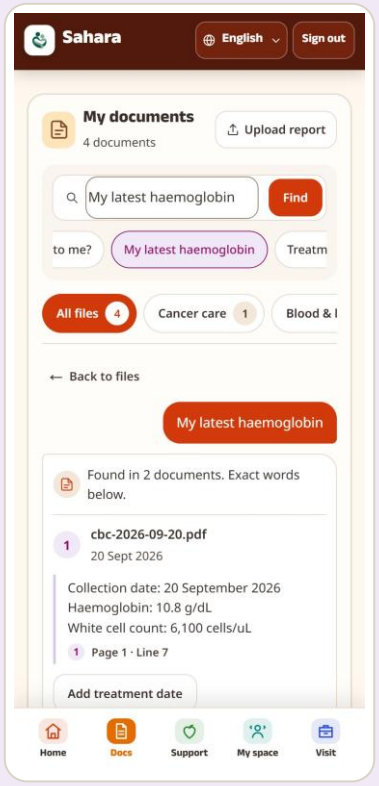
Doctor and patient or family sign; it locks and opens from a QR code in an emergency

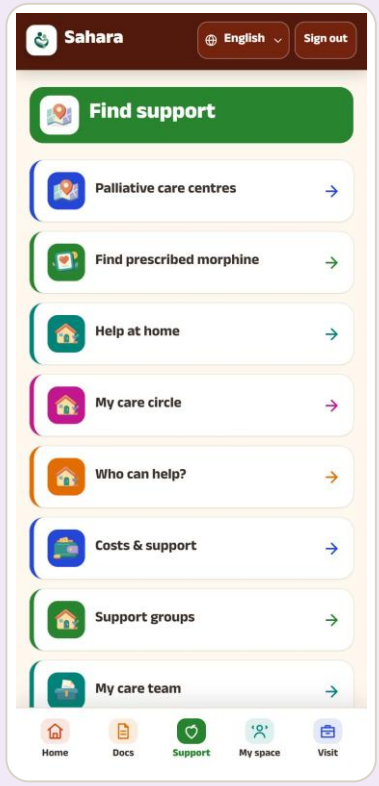
Solution Overview, Key Features & Impact (2/2)

At home, between visits: patients and families log how things are going; their doctor sees the trends.

Every feature is built on published guideline tools and validated questionnaires (ESAS-r symptoms, NCCN Distress Thermometer, ZBI-4 caregiver burden).







- **How I feel:** symptoms (ESAS-r), a 0–10 distress scale (NCCN thermometer) and a pain body map; trends for the doctor.
- **Caregiver wellbeing (new):** ZBI-4 check and a private journal; flagged when it worsens, with Tele-MANAS 14416, counsellors, respite care, support groups.
- **Medicines:** pill tracker from prescription photos; tag pictures of your medicine strips for easy identification; dose ticks, reminders.
- **Documents:** uploads sorted into prescriptions, bills, reports; questions answered with the line it came from.
- **Emergency:** SOS button and a QR code with goals of care, contacts and medicines.
- **Help nearby:** palliative-care centres and availability of morphine, emergency hospitals, ambulances, support groups.
- **Family:** invited relatives share the record with their own access level; ability to manage caregiver concerns.

What we expect to change (to be measured in a pilot):

more of these talks written and signed · wishes found in seconds in an emergency · fewer family arguments at the bedside · doctors see problems before the next visit.

Technology, Tools, and AI Approach

How a conversation becomes a signed clinical document

Doctor's phone
records only after patient consent

→

Our server
keeps the AI password; the phone never has it

→

Google Gemini AI
types out the talk and fills in the clinical document

→

Doctor
reads it, corrects it, signs; the patient or caregiver also signs

→

Locked document
PDF + QR code; any later change creates a new version that requires re-signatures

No internet, or the AI is busy? OpenAI Whisper turns speech into text on the phone itself. If that fails too, the doctor types.

Built with

Android
the phone app

Kotlin + Compose
language and screens of the app

Next.js
the website

Vercel
puts the website online

Supabase
server, sign-in, storage

PostgreSQL
database; each family sees only its own

Google Gemini
AI: speech to text, fills the document

OpenAI Whisper
speech to text with no internet

Google ML Kit
5 languages and read-aloud on the phone

OpenStreetMap
free map of nearby centres

Clinical tools
ESAS-r, NCCN Distress Thermometer, ZBI-4

FHIR
standard format hospital software can read

Build Sprint Journey and Progress

October – November 2026

Built · Doctor / legal feedback · Milestone

	MON	TUE	WED	THU	FRI	SAT	SUN
Week 1	5 Oct	6	7	8	9 Shortlisted! Android app + map of centres	10 AI medicine organiser	11 Prescription organiser
Week 2	12	13 Google Play test version	14 Testers invited	15	16 Privacy and account safety	17	18
Week 3	19 Sent to Google Play	20 Mentor review	21	22	23 Hospital survey feedback	24	25 Symptom tracker
Week 4	26 Pain body map	27 Hospital survey live	28 Team meeting: 9 changes Ask your reports	29 Dr Sharada's medical rules	30 Goals of care + QR code Telugu, Urdu, Kannada	31 Legal advice on format	1 Nov Recording → text
	2 Doctor schedule	3 Dr Sujay tests the whole app	4 Clinical document word for word 8 practice talks pass	5 Gemini AI + backups	6 Calmer, fewer taps	7 Dr Sujay's 13 topics Caregiver wellbeing	8

Blue circle: 7 Nov, the day of this submission. Grey days are outside the build sprint.

✓ Working now

Patient and family app · doctor's view · recording → signed clinical document · QR code · website live

↻ Changed on the way

Website first, then a real Android app. The AI used to write the note; now the doctor leads and the AI only writes it down.

⚠ Problems we solved

Google's AI was often busy → automatic backups. Exact goals-of-care wording → copied.

… Still to do

Test on many real phones · record a real practice talk with Dr Sharada

Evaluation and User Validation

1. Does it work? Our own tests

877
automatic checks on the Android app: all pass

594 / 594
automatic checks on the website: all pass

8 / 8
practice doctor–patient talks in 6 languages: every answer landed in the right place on the document

4 / 4
questions about a sample hospital report answered correctly

Automatic checks are small test programs that run after every change, so a broken feature is caught before release. The practice talks are made up, not real patients.

2. What experts told us, and what we changed

Dr Sujay Halkur Shankar
doctor, used the whole app on 3 Nov

Said: Is there a guide telling the doctor what to talk about next? And a medicine taken 4 times a day needs 4 ticks.

We changed: Checklist in the doctors' order with a "not yet discussed" prompt; one tick per dose. Team chat: tubes-and-drains tracker, darker text.

Dr Sharada Vinod Kutty
doctor, wrote our medical rules on 29 Oct

Said: Comfort care must always continue, whatever treatment a patient refuses.

We changed: The app shows this on every signed document.

Not tested yet: with real patients · on many real phones · speech-to-text on real Hindi, Telugu and Kannada recordings · a doctor checking each translation means the same

Safety and Responsible AI

Privacy and consent

- ✓ Recording starts only after the patient ticks “I agree”
- ✓ The AI is used only with that consent
- ✓ Each family sees only their own records

Limits on the AI

- ✓ The AI never suggests treatment
- ✓ If a topic was not discussed, it says so; it does not guess
- ✓ If the AI is down, the phone or the doctor writes it

A person decides

- ✓ The doctor reads, corrects and signs first
- ✓ The patient or family signs too
- ✓ Any later change creates a new version requiring re-signatures

For everyone

- ✓ 5 Indian languages, and it reads text aloud
- ✓ Sign with a thumbprint; large text
- ✓ Demos use made-up patients only

Deployment, Cost and Sustainability Plan

Pilot
Start in 1 clinic with our own doctors. Patients give consent as India's data-protection law (DPDP Act) requires.

Fitting in
Today it records on the doctor's phone. Next: ambient listening, with room or lapel microphones in clinics. Later, FHIR export for hospital software.

Who pays
Clinics or charities pay a monthly fee. Free for patients and families.

Growing
Grants first, then palliative-care networks across India.

What it costs	Once	Every month
Servers and website (Supabase + Vercel)	–	₹4,200
Google Gemini AI, per 30-minute talk	–	≈ ₹2 a talk
Setting up and training clinic staff	₹25,000	–
Consent forms and legal check	₹25,000	–
Support from our team	–	₹10,000
Cost per clinic, if 10 clinics share it	**₹5,000**	**≈ ₹1,600**

We would charge ₹1,500–2,000 per clinic a month

Our estimates from public prices: US$1 ≈ ₹94; Supabase $25 and Vercel $20 a month; Gemini ≈ $0.30 per million audio units (a 30-minute talk ≈ 58,000); 100 talks per clinic a month.

What Next? Plans to take initiative forward (if any)

1

Prove it works

Test on many real phones; try room and lapel microphones (ambient listening). Record a real practice talk with Dr Sharada.

2

Pilot, 60–90 days

One clinic. Count how many talks get signed and how fast wishes are found.

3

Legal rules

What to do when a patient cannot decide, who may sign for them, and how a patient changes their goals of care.

4

Fund and grow

Grants, palliative-care networks, then link with hospital software.

Next version of the app: more features such as devices like feeding tubes · an improved caregiver portal · updates based on user feedback

Can the organisers share our solution publicly? YES

Slides, brief, video and app description. Not patient records, passwords or private code.

Health-a-thon 2026

Thank you

Zeros and Ones • 37MYUP

Try it: sahara.wedevit.in | Demo video: [add demo video link] | workwithdevit@gmail.com | Code: github.com/basithladdu/healthathon (private)

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